DIFFDOCK: DIFFUSION STEPS, TWISTS, AND TURNS FOR MOLECULAR DOCKING

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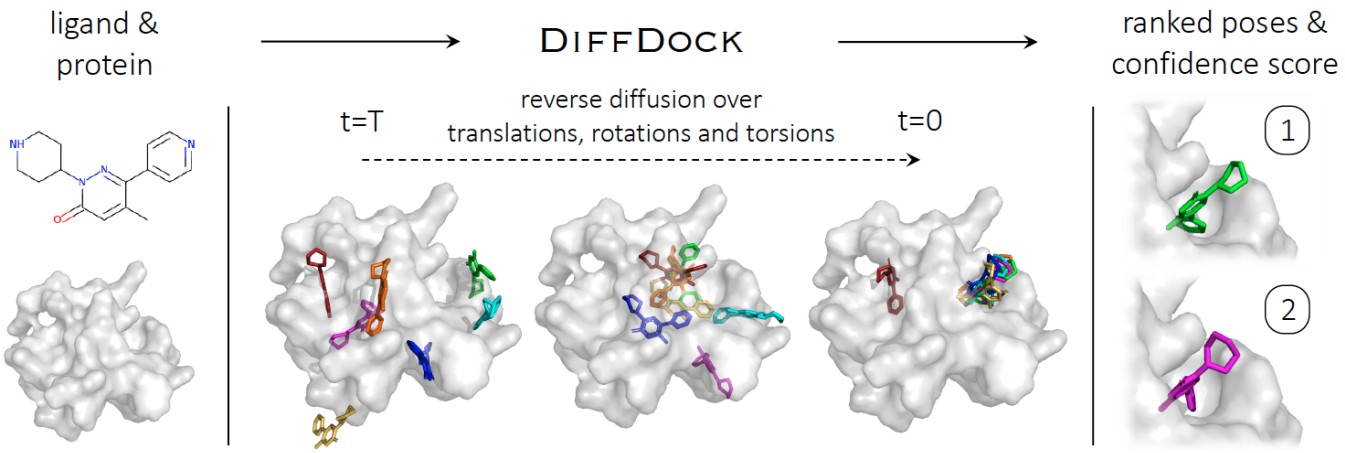


Figure 1: Seattle Mariners at Spring Training, 2010.

Keywords

Ligand, Molecular Docking

1 Introduction

1.1 Overview of Molecular Docking

Molecular docking is a computational technique used to predict the interaction between a small molecule, ligand, and a protein. This process is central to drug discovery, where the binding structure of ligands to proteins can determine the therapeutic effect of the drug. The challenge lies in accurately modeling the ligand's binding pose, which includes its position, orientation, and conformation within the protein's binding site. Docking plays a critical role in virtual screening pipelines, where thousands of candidates are evaluated for potential efficacy.

1.2 Challenges in Ligand-Protein Biding Prediction

Docking is really complex due to the large search and multimodal nature of binding poses. Traditional methods rely on scoring functions and search algorithms, which are computationnaly expensive and often inaccurate in capturing the full diversity of valid poses. While deep learning has accelerated docking predictions, existing methods typically frame docking as a regression task, predicting a single binding pose. This approach fails to account for multiple plausible binding configurations (multimodal distributions), leading to suboptimal predictions.

1.3 Framing DiffDock as a Generative Problem

The goal of DiffDock is to approach molecular docking as a generative problem. DiffDock learns the distribution of possible ligand poses—including their positions, orientations, torsion, and angles—allowing it to handle multiple plausible binding configurations effectively.
DiffDock reframes the classical problem in molecular docking as a generative modeling problem. Unlike regression-based methods, it uses a diffusion generative model (DGM) to predict a distribution over possible ligand poses. This approach enables DiffDock to efficiently sample multiple plausible binding poses while maintaining accuracy and confidence scores. The model operates on a reduced manifold of ligand poses, leveraging ligand-specific degrees of freedom, such as translations, rotations, and torsion angles, to simplify computations and improve performance.
The task is to generate realistic ligand poses (positions, orientations, and torsional conformations) that fit into a protein's binding pocket

1.4 Objectives and Structure of This Report

This report aims to present and explain how DiffDock works, focusing on the innovative methodology and key contributions of the paper. We will also provide a critical perspective of the paper, highlighting its strengths, limitations, and potential improvements. The report is structured as follows:

- Section 2 places DiffDock in the context of related work and its contributions to molecular docking.
- Section 3 digs into the methodology, explaining the diffusion generative model and the transformation to a simplified product space.
- Section 4 evaluates the performance of Diffdock and its advantages over traditional deep learning based method.
- Section 5 offer a critical analysis, discussing the model's strenghts, limitations and areas of improvement.
- Section 6 details an attempt to reproduce key experiments and explore modifications. Finally, the report concludes with a summary of findings, broader implications for drug discovery, and suggestions for future research directions

2 Context and Related Work

2.1 Traditional Docking Methods

Traditional docking methods rely on optimizing a scoring function to predict the most favorable ligand pose. Examples include well-established methods such as AutoDock Vina and SMINA. These methods evaluate ligand-protein interactions and search for the pose with the best energy score.

Challenges:

- **Computational Cost:** The optimization process is computationally expensive, often requiring significant time for exhaustive searches.
- **Sensitivity to Initial Conditions:** The results are heavily influenced by the initial poses provided, leading to potential inaccuracies.
- **Noisy Data:** These methods struggle with noisy or incomplete input data, affecting their robustness and reliability.

2.2 Deep Learning in Docking

Deep learning-based methods have introduced data-driven approaches to docking. Techniques like EquiBind and TANKBind predict ligand poses using regression-based models.

Advantages:

- **Speed:** These methods are significantly faster than traditional search-based docking.

Limitations:

- **Accuracy:** Deep learning models often fail to achieve the accuracy of traditional methods. This is largely due to their inability to:
  (1) Represent multimodal distributions, which are common in docking tasks.
  (2) Account for uncertainties, as regression methods tend to average over possible solutions.

2.3 Generative Models for Multimodal Distributions

Generative models, especially DGMs, have demonstrated success in tasks requiring multimodal output distributions. These models are particularly effective in molecular machine learning tasks such as:

- Molecule generation,
- Protein structure design.

Diffusion Generative Models:

- DGMs work by progressively refining noise into structured outputs, making them well-suited for generating diverse and accurate solutions.
- Their ability to handle multimodal distributions is a significant advantage over deterministic regression methods.

2.4 DiffDock's Contributions in Context

DiffDock adapts DGMs to the docking problem by defining a diffusion process over the ligand pose space. This novel approach addresses key limitations of previous methods:

(1) **Multimodal Distribution Representation:** DiffDock can model the diverse set of possible ligand poses.
(2) **Uncertainty Modeling:** The diffusion framework inherently accounts for uncertainties, improving robustness.
(3) **Efficiency:** By leveraging the product space transformation, DiffDock ensures computational efficiency while maintaining accuracy.

Key Innovations:

- **Transformation to Product Space:** DiffDock redefines the problem in a space where diffusion is more computationally feasible.
- **Generative Modeling:** The method shifts from deterministic predictions to probabilistic sampling, capturing the inherent complexities of ligand-protein interactions.

Impact: DiffDock represents a significant advancement in molecular docking, combining the strengths of generative modeling with domain-specific adaptations to achieve state-of-the-art results.

3 Methodology

3.1 Too Many Possibilities: Degrees of Freedom in Molecular Docking

The pose of a ligand corresponds to the atomic positions of its atoms in 3D space, which can be mathematically represented as:

$$x \in \mathbb{R}^{3n}$$

where n is the number of atoms in the ligand, and x is a vector containing the coordinates of all n atoms. For instance,

a ligand with four atoms can be represented as:

$$x = \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \\ x_4 & y_4 & z_4 \end{bmatrix}$$

where (x_i, y_i, z_i) represents the 3D position of the i-th atom.

However, working directly in the high-dimensional space $\mathbb{R}^{3n}$ introduces unnecessary complexity, as it assumes complete freedom for every atomic position. In reality, ligand flexibility is significantly constrained by physical and chemical properties:

- **Bond lengths:** Fixed distances between bonded atoms.
- **Bond angles:** Fixed angles between bonds.
- **Small rings:** Rigid due to chemical constraints.

The primary source of flexibility in ligands comes from **torsion angles**, which describe rotations around rotatable bonds. Thus, instead of modeling ligand poses in the full $\mathbb{R}^{3n}$, the problem can be simplified to a lower-dimensional $(m+6)$-dimensional manifold, where:

$$m = \text{Number of rotatable bonds.}$$

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3.2 Transformation to Product Space

The idea of the product space comes from the following assumption:

> Any ligand pose consistent with a seed conformation can be reached by a combination of (1) ligand translations, (2) ligand rotations, and (3) changes to torsion angles.

3.2.1 Translation, Rotation, and Torsion Spaces. DiffDock maps $\mathcal{M}_c$ to a product space, simplifying the diffusion process:

$$P = T(3) \times SO(3) \times SO(2)^m$$

where:

(1) $T(3)$: Translation in 3D space. The translation operation is defined as:

$$A_{\text{tr}} : T(3) \times \mathbb{R}^{3n} \to \mathbb{R}^{3n}$$

$$A_{\text{tr}}(r, x)_i = x_i + r,$$

where $r \in T(3)$ is the translation vector, and x_i represents the position of the i-th atom.

(2) $SO(3)$: Rotation in 3D space. The rotation operation is defined as:

$$A_{\text{rot}} : SO(3) \times \mathbb{R}^{3n} \to \mathbb{R}^{3n}$$

$$A_{\text{rot}}(R, x)_i = R(x_i - \bar{x}) + \bar{x},$$

where $R \in SO(3)$ is the rotation matrix, and $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ is the center of mass of the ligand.

(3) $SO(2)^m$: Torsion space for m rotatable bonds. The torsion operation is defined as:

$$A_{\text{tor}} : SO(2)^m \times \mathbb{R}^{3n} \to \mathbb{R}^{3n}.$$

Each torsion update rotates parts of the ligand around a rotatable bond while preserving bond lengths and angles.

3.2.2 Simplification via RMSD Alignment. To ensure that transformations in P do not interfere with each other, torsion updates are simplified using RMSD alignment.

Definition of Torsion Updates. Let $x \in \mathbb{R}^{3n}$ represent the positions of the n atoms in the ligand. For the k-th rotatable bond (a_k, b_k), a torsion update is denoted as:

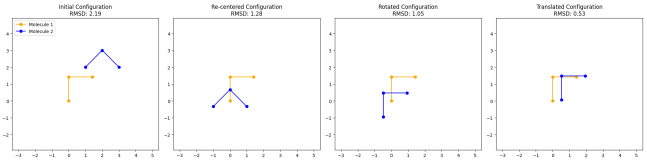
$$B_{k,\theta_k}(x),$$

where θ_k is the torsion angle. The combined torsion update for all m rotatable bonds is given by:

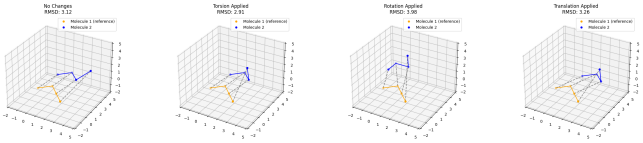
$$(B_{1,\theta_1} \circ B_{2,\theta_2} \circ \ldots \circ B_{m,\theta_m})(x).$$

After applying the torsion updates, the ligand might be slightly displaced in 3D space. To correct this, the updated structure is aligned back to the original pose by minimizing the **Root Mean Square Deviation (RMSD)**:

$$\text{RMSD}(x, y) = \sqrt{\frac{1}{n}\sum_{i=1}^{n} \|x_i - y_i\|^2}.$$



(a) 2D visualization of RMSD minimization stages, showing the alignment process of two molecules (Molecule 1 and Molecule 2) through re-centering, rotation, and translation.



(b) 3D visualization of RMSD and structural changes, including torsion, rotation, and translation for Molecule 2 relative to Molecule 1 (reference molecule).

Figure 2: Comparative analysis of RMSD minimization and structural transformations in both 2D and 3D space. The diagrams highlight how alignment techniques improve molecular correspondence.

Definition of RMSD Alignment. RMSD alignment is a function that minimizes the RMSD by finding the optimal rotation and translation of the updated structure x' to align with the original structure x:

$$\text{RMSDAlign}(x, x') = \text{argmin}_{x^\dagger \in \{gx' | g \in SE(3)\}} \text{RMSD}(x, x^\dagger),$$

where $SE(3)$ represents the group of all rigid-body transformations (translations and rotations).

The torsion operation A_{tor} is therefore defined as:

$$A_{\text{tor}}(x, \theta) = \text{RMSDAlign}\left(x, (B_{1,\theta_1} \circ \cdots \circ B_{m,\theta_m})(x)\right).$$

Combining Transformations. Finally, a complete ligand pose transformation is expressed as:

$$A((r, R, \theta), x) = A_{\text{tr}}(r, A_{\text{rot}}(R, A_{\text{tor}}(\theta, x))).$$

This equation encapsulates all the degrees of freedom involved in docking.

And so we can formally define space of ligand poses as:

$$\mathcal{M}_c = \{A(g, c) | g \in P, c \in \mathbb{R}^{3n}\}$$

3.3 Defining the Diffusion Process on the Product Space

The paper establishes that the function $A(\cdot, c) : P \to \mathcal{M}_c$ is a bijection (Proposition 2). This bijection ensures that every point in the product space $P = T(3) \times SO(3) \times SO(2)^m$ corresponds to a unique ligand pose in $\mathcal{M}_c$. Consequently, we can define the diffusion process directly in P and train the model to learn the distribution of ligand poses effectively [2].

3.4 Diffusion on the Product Space

This section explains how the diffusion process is defined on the product space $P = T(3) \times SO(3) \times SO(2)^m$, representing the degrees of freedom in ligand docking. By leveraging the properties of P, the generative model avoids the complexities of operating directly on the manifold $\mathcal{M}_c$ of ligand poses.

3.4.1 Mapping to the Product Space. Proposition 2 proves that the map $A(\cdot, c) : P \to \mathcal{M}_c$ is a bijection. Thus, the diffusion process can be formulated on P, where the sampling and score computations are computationally efficient.

3.4.2 Defining the Diffusion Process. The diffusion process progressively adds noise to ligand poses in P, transforming them into random configurations. This process is governed by the forward stochastic differential equation (SDE):

$$dx = \sqrt{\frac{d\sigma^2(t)}{dt}}\, dw$$

where $\sigma^2(t)$ is the variance of the noise, and w is a Brownian motion associated with each group.

The reverse diffusion process refines a noisy pose into a valid ligand pose:

$$dx = \nabla \log p_t(x)\, dt + g(t) dw$$

where $\nabla \log p_t(x)$ is the score function that guides the denoising process.

Translation Group ($T(3)$). For translations, $T(3) \cong \mathbb{R}^3$, so the forward SDE corresponds to standard Brownian motion:

$$dx = \sqrt{\frac{d\sigma_{\text{tr}}^2(t)}{dt}}\, dw$$

where $w \sim \mathcal{N}(0, \sigma_{\text{tr}}^2(t))$. Intuitively, this models random drifting of the ligand in 3D space.

Rotation Group ($SO(3)$). For rotations, the diffusion kernel is described by the IGSO(3) distribution:

$$p(\omega) = \frac{1 - \cos\omega}{\pi} f(\omega), \quad f(\omega) = \sum_{l=0}^{\infty} (2l+1) e^{-l(l+1)\sigma^2/2} \frac{\sin((l + \frac{1}{2})\omega)}{\sin(\omega/2)}$$

where ω is the angle of rotation. The score of the kernel is:

$$\nabla \ln p_t(R'|R) = \left(\frac{d}{d\omega} \log f(\omega)\right) \hat{\omega}$$

Sampling and score computation are made efficient by precomputing $f(\omega)$ as a truncated series and interpolating its cumulative distribution function (CDF).

Torsion Group ($SO(2)^m$). For torsions, $SO(2)^m$ is diffeomorphic to the torus T^m. The diffusion kernel is modeled as a wrapped normal distribution:

$$p_t(\theta) \sim \text{Wrapped Normal}(\theta; \mu, \sigma_{\text{tor}}^2(t))$$

where θ are the torsion angles. Sampling is straightforward, and the score is precomputed as a truncated series.

3.5 Training and Inference

Althought the diffusion process is mathematically defined in the product space P, the trianing and inference procedure are implemented in 3D space.

3.5.1 Why 3D Space instead of P?. This is much due to physical constraint. Ligands intereact ith proteins in 3D space, where phhysical constraints (like steric clashes or hydrogen bonds) are more easily defined.
By training the model in 3D space, the model leverage $SE(3)$ equivariance, which is important because intereactions between ligands and proteins don't depend on the absolute position of the ligand, but rather on the relative position between the ligand and the protein.

They also claim in their paper that torsion angles depend on how bond are parametrized and so directly training on 3D avoids the arbitrary choices, improving generalization.

Another thing that they try to minimize is the dependence on the choice of the seed conformation. By training in 3D space, the model is less dependent on the seed conformation, which is a good thing because the seed conformation is often arbitrary and can be a source of bias.

3.5.2 Confidence model. Once the diffusion model has generated a ligand pose, a confidence model is used to estimate the quality of the pose.

How is it trained? First we need to generate training data. The trained diffusion model is used to produce candidate poses for each example in the training set.
Each candidate pose is labeled as valid or invalid based on its RMSD is below a certain threshold (2Å in the paper).
RMSD $<$ 2Å is considered a valid pose, while RMSD *geq* 2Å is considered invalid.

Then the confidence model is trained to predict the probability that a candidate pose is valid using cross-entropy loss.

How does the confidence model work during inference? The diffusion model is run N times in parallel to produce a set of candidate poses for the ligand.
Then each candidate pose is passed to the confidence model,which outputs a confidence score indicating its likelihood of being a valid pose.
And then the poses are ranked based on their confidence scores, and the top-ranked pose is selected as the final prediction.

3.6 Model Architecture

The proposed framework comprises two key components: the score model $s(x, y, t)$ and the confidence model $d(x, y)$. Both operate on 3D representations of the ligand pose x and the protein structure y, capturing the physical and geometric features of the molecular system.

The score model predicts the gradients of the diffusion process required to refine ligand poses. Specifically, it outputs two SE(3)-equivariant vectors: a translation vector $\mathbb{R}^3$ for adjusting the ligand's position and a rotation vector $\mathbb{R}^3$ for orienting the ligand. Additionally, it predicts an SE(3)-invariant scalar for each rotatable bond, corresponding to torsion adjustments in $\mathbb{R}^m$. This design ensures the model respects the geometric symmetries inherent to molecular systems while enabling effective refinement of ligand poses.

To achieve computational efficiency, the score model operates on a coarse-grained representation of the protein, considering only α-carbon atoms. This simplification retains sufficient geometric detail for pose refinement while significantly reducing the computational cost.

The confidence model complements the score model by ranking candidate poses based on their likelihood of being physically valid, as measured by the RMSD. It outputs a single SE(3)-invariant scalar that evaluates the quality of each pose. Unlike the score model, the confidence model processes the full atomic structure of the protein, leveraging detailed features for precise scoring. The ligand atom features are mean-pooled, and the resulting representation is passed through a fully connected layer to produce the scalar confidence score.

Both models leverage a geometric graph-based representation of the molecular structures, where nodes correspond to ligand atoms and protein residues (or atoms in the case of the confidence model) and edges encode spatial relationships. Residue nodes are initialized with features derived from protein sequence embeddings trained using language models, enhancing the representation with contextual biological information.

To compute the outputs, translational and rotational scores are derived by convolving node features with a tensor product filter positioned at the ligand's center of mass. Torsional scores are computed using a pseudotorque convolution, adapted to the coarse-grained representation of the protein. This multiscale architecture balances physical accuracy and computational efficiency, enabling robust predictions across diverse ligand-protein complexes.

4 Results and Analysis

4.1 Quantitative Performance on PDBBind

DiffDock was evaluated on the PDBBind benchmark, a widely used dataset for molecular docking, and demonstrated significant improvements in performance. The top-1 success rate was 38.2% when sampling 40 poses and 35.0% for 10 poses, outperforming the previous state-of-the-art methods. Notably, DiffDock surpassed traditional search-based methods such as GLIDE (21.8%) and recent deep learning approaches like TANKBind (20.4%). These results underscore DiffDock's ability to efficiently predict accurate ligand poses.

4.2 Robustness to Folded Protein Structures

DiffDock exhibited robustness in docking with computationally folded protein structures, such as those generated by ESMFold. Traditional methods often struggle in this scenario, achieving maximum accuracies around 10.4%. In contrast, DiffDock maintained a top-1 success rate of 22% for apo-structures without additional training. This robustness is attributed to the diffusion model's flexibility in handling small structural perturbations and its ability to accommodate imperfections in side-chain conformations.

4.3 Comparison with Traditional and Deep Learning Methods

DiffDock was compared against a variety of existing methods:

- **Traditional search-based methods:** These include SMINA, GLIDE, and GNINA. While these methods perform adequately with holo-structures, their accuracy decreases significantly for apo-structures. DiffDock outperformed these methods in both speed and accuracy.
- **Deep learning methods:** EquiBind and TANKBind are recent ML-based docking approaches. DiffDock achieved nearly double the success rate of these models and provided diverse pose predictions. Unlike regression-based models, DiffDock handles uncertain binding scenarios effectively by generating multiple plausible poses.

4.4 Visual Examples of Predicted Ligand Poses

To gain a deeper understanding of how DiffDock operates, we applied the model to the molecule 4-methoxybenzonitrile (COc(cc1)ccc1C#N) 3, whose structure is illustrated below. This ligand was docked with the protein 6agt, which represents the crystal structure of PfKRS (Plasmodium falciparum lysyl-tRNA synthetase) in complex with a chromone inhibitor.

To illustrate the performance of DiffDock, we present visual examples of predicted ligand poses within a protein binding pocket and their corresponding evaluation metrics. The following figures highlight a top-ranked pose (assigned the highest confidence score by DiffDock), a non-top-ranked pose, and the relationship between DiffDock confidence scores and GNINA binding affinity predictions.

The model was tested using the following hyperparameters: 20 inference steps, 40 samples generated per protein-ligand complex, a batch size of 6.

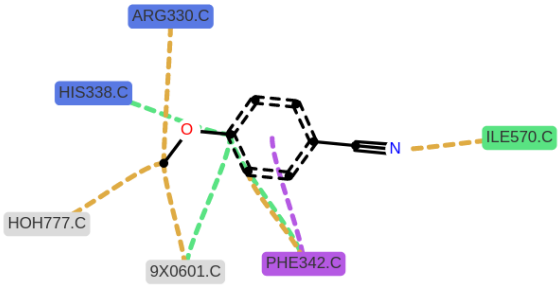


Figure 3: Molecular structure of 4-methoxybenzonitrile (COc(cc1)ccc1C#N), a ligand used in the DiffDock evaluation.

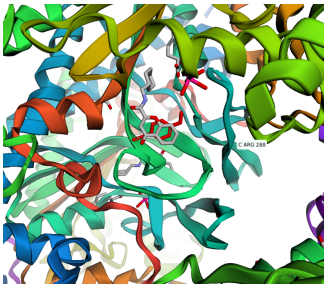


Figure 4: Top-ranked ligand pose predicted by DiffDock. The red-colored ligand is positioned within the protein's binding pocket, forming strong interactions with surrounding residues (colored helices). This pose exhibits high alignment with the experimental binding conformation and likely achieves a low RMSD, reflected in its high DiffDock confidence score.

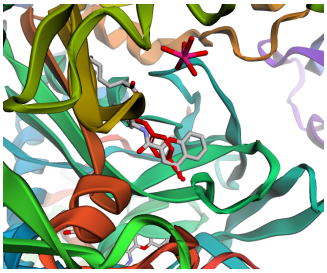


Figure 5: Non-top-ranked (top 10) ligand pose predicted by DiffDock. The ligand (red) exhibits weaker alignment and fewer interactions with the binding pocket.

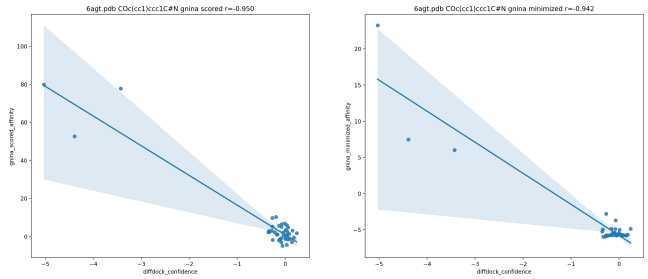


Figure 6: Correlation between DiffDock confidence scores and GNINA binding affinity predictions. The left plot shows GNINA-scored affinities versus DiffDock confidence scores, with a strong negative correlation ($r = -0.950$). The right plot shows minimized affinities from GNINA versus DiffDock confidence scores, also demonstrating a strong negative correlation ($r = -0.942$). These results validate the effectiveness of DiffDock's confidence model in ranking ligand poses, as higher confidence scores correspond to stronger predicted binding affinities.

These visual examples highlight DiffDock's ability to generate accurate and diverse ligand poses while effectively ranking them based on predicted binding affinities. The top-ranked pose demonstrates strong geometric and interaction alignment, while the correlation analysis confirms the reliability of DiffDock's confidence scoring mechanism.

5 Critical Perspective

5.1 Strengths of DiffDock

5.1.1 Handling Multimodal Binding Poses. DiffDock's generative approach allows it to predict multiple plausible binding poses for a single ligand, addressing the inherent uncertainty in molecular docking. Unlike regression-based methods that may converge to incorrect local optima, DiffDock's ability to sample diverse poses captures the multimodal nature of ligand-protein interactions. This is particularly beneficial for systems with ambiguous or flexible binding pockets, where traditional methods often fail. It seems to be a quite novel approach to the problem of molecular docking.

5.1.2 Confidence Scoring and Practicality. The inclusion of a confidence model is a significant practical advantage. By assigning a confidence score to each predicted pose, DiffDock enables users to rank and filter poses based on their likelihood of being physically accurate. This feature integrates seamlessly with downstream workflows, such as experimental validation or further computational analysis, enhancing the practical utility of the model in drug discovery pipelines.

5.2 Limitations and Shortcomings

5.2.1 Synthesizability of Predicted Ligands. While DiffDock excels at predicting plausible binding poses, it does not account for the synthesizability of ligands. Predicted poses may involve conformations or chemical modifications that are challenging or impossible to achieve in practice. This limitation could hinder the transition from computational predictions to experimental validation, particularly in early-stage drug design.

5.2.2 Computational Complexity. Although DiffDock's coarse-grained representations improve efficiency, its diffusion-based approach remains computationally intensive compared to traditional docking methods. For large-scale applications involving thousands of protein-ligand pairs, this complexity could become a bottleneck. Moreover, the requirement to sample multiple poses increases the computational cost further, potentially limiting its use in resource-constrained settings.

5.2.3 No error bars. Another issue that is quite surprising is the lack of error bars. The model is able to predict the confidence of a pose, but it doesn't provide any uncertainty estimate. This is a major drawback. It doesn't seems to have a table or graph showing confidence intervals or standard deviations. This is a major drawback because it's important to know how confident the model is in its predictions.

5.2.4 Only α-carbon atoms are considered. It seems that only α-carbon atoms are considered in the model. This is a major limitation because it's known that the side chains of amino acids are important for ligand binding. This could be a major source of error in the model.

5.2.5 No explanation about the preprocessing of the data. It's not very clear how the data was preprocessed. It would be interesting to know how the data was preprocessed before being fed to the model. This is important because the quality of the data can have a major impact on the performance of the model. For example when evaluating the code, if raw structures were used as input to the other docking methods, then it is no surprise they performed poorly. The authors should describe the preprocessing steps in more detail.

5.3 Suggestions for Improvement

5.3.1 Incorporating Chemical Feasibility: One significant gap in DiffDock is the lack of consideration for ligand synthesizability. Integrating cheminformatics tools or methods like SynFlowNet [3], which uses reinforcement learning to generate chemically feasible molecules from small building blocks, could help ensure that the predicted ligands are experimentally realizable. Similar efforts, such as REINVENT [9], focus on reinforcement learning for generating synthesizable drug candidates, making them promising frameworks for incorporation.

5.3.2 Improving Efficiency: Diffusion-based models, while powerful, remain computationally demanding. Techniques like shared diffusion kernels or model distillation could significantly improve sampling efficiency. For instance, Score-Based Generative Models with Distillation [8] have demonstrated how distillation can reduce the number of sampling steps required in generative processes without losing accuracy. Leveraging such techniques could reduce the computational overhead of DiffDock, making it more scalable.

5.3.3 Dynamic Protein Structures: Many proteins exhibit conformational flexibility that is not accounted for in static docking models. Efforts like AlphaFold-Multimer [5] and ensemble-based docking approaches [1] demonstrate the value of capturing multiple protein conformations. Adapting DiffDock to operate on ensembles of protein structures could significantly improve its applicability to targets with high structural variability.

5.3.4 Integration with Virtual Screening Pipelines: To enhance its utility, DiffDock could be coupled with virtual screening tools that generate ligand libraries, like DeepChem [7] or generative adversarial network-based ligand designers such as MolGAN [4]. This integration would create an end-to-end pipeline for ligand generation, docking, and evaluation, streamlining early-stage drug discovery workflows.

6 Reproduction of Results and Modifications

6.1 Challenges in Reproducing Results

Reproducing the DiffDock results on the full dataset was not feasible due to computational limitations. Using Google Colab with restricted GPU resources, we focused on testing the model with a smaller subset of ligands rather than the full dataset. While this allowed us to evaluate certain aspects of the method, the lack of sufficient computational power prevented a complete reproduction of the original experiments.

6.2 Observations and Insights from Reproduced Results

Two experiments were conducted using different hyperparameter settings to evaluate DiffDock's ability to generate and rank ligand poses. The first experiment utilized 50 inference steps, 20 samples per complex, a batch size of 12, and a cosine sigma schedule, while the second experiment used 20 inference steps, 40 samples per complex, and a batch size of 6. Both experiments were performed on the same protein-ligand pair (6agt.pdb and COc(cc1)ccc1C#N), but the rankings and confidence scores for the poses were significantly different.

6.2.1 Observations. **Ranking Sensitivity to Hyperparameters:** The first experiment, with a higher number of inference steps (50) and fewer samples (20), led to a more significant reshuffling of the ranking order for poses with lower DiffDock confidence scores. This suggests that increasing inference steps improves pose refinement, leading to greater divergence in ranking based on confidence scores.

Pose Stability at the Top: Despite the hyperparameter changes, the top pose remained the same in both experiments, achieving a DiffDock confidence score of 0.24. This indicates that the highest-scoring poses are robust to changes in inference steps and sample size.

Impact of Sample Diversity: The second experiment generated more samples per complex (40 vs. 20), likely increasing the diversity of poses but possibly reducing the refinement quality due to fewer inference steps (20 vs. 50). This may explain the observed differences in ranking for lower-confidence poses.

Confidence Score Interpretation: The first experiment resulted in a broader range of confidence scores for lower-ranked poses, reflecting more refined differentiation. In contrast, the second experiment exhibited smaller differences in scores, suggesting less effective refinement with fewer inference steps.

6.2.2 An experiment we tried but failed. One thing we have tried is to investigate we DiffDock could work with some proteins and ligand that we supposed to be easy to dock as we can see on Fig 7.

Case	Protein (PDB)	Ligand (SMILES)	DiffDock Confidence	GNINA Scored Affinity	GNINA Minimized Affinity
Flexible Binding Site	HIV Protease (1HVR)	`CC(C)(C)OC(=O)N1CCC(O1=O)O`	-0.48	-4.67633	-5.04885
Covalent Binding	EGFR (5UG8)	`C1CC=NC2=C(C=C(C=C2)NC3=CC=C(C=C3)C(F)(F)F`	-0.60	-6.26226	-7.83974
Metal Coordination	Carbonic Anhydrase II (3KS3)	`CC1=NN=C(S1)S(=O)(=O)N`	0.95	-5.71000	-6.18093
Baseline Case (Proposed)	Sirtuin (6AGT)	`CDc(c1)ccc1C#N`	0.22	-0.05776	-5.60777

Figure 7: Summary of the proteins and ligands used in the experiment to test DiffDock's performance with some proteins and ligand with different properties like.

What we were trying to show, is to investigate our hypothesis that DiffDock would perform well with easy cases. We chose some proteins and ligands that are known to be easy to dock, and we tried to see if DiffDock would perform well with them. And some others that are far from being easy to dock and far from what the model have seen during the training, as explained in the limitations section we were quiet doubtful about the model's ability to generalize to unseen data.

We failed to find any relevant information that might confirm this hypothesis.

But 3 Dec 2024, a paper was published that showed that DiffDock performed well on easy cases, but not on hard cases [6]. The paper critically evaluates DiffDock and compares it to conventional docking approaches such as Surflex-Dock, Glide, AutoDock Vina, and Gnina. The study highlights that DiffDock's reported superior performance stems from a heavy reliance on training data with near-identical protein-ligand pairs, which artificially inflates its success rates. When tested on more challenging cases without similar training data, DiffDock's performance significantly declines.

The authors established a fair comparison by using a clean test set and fully automated workflows for traditional methods. Results showed that conventional docking methods outperformed DiffDock, especially in scenarios where the binding site was predefined. Even in blind docking scenarios, where the binding site was unknown, traditional methods demonstrated better reliability and accuracy.

DiffDock's approach was criticized for relying on a heavily imbalanced 98/2% training-testing split, which favors memorization over generalization. This undermines its applicability to real-world drug discovery challenges, where novel protein-ligand interactions are the primary focus. In contrast, conventional methods showed more consistent performance across diverse scenarios.

The authors concluded that while machine learning has potential in molecular docking, its practical applications must be rigorously validated. They called for careful benchmarking and emphasized the importance of addressing challenges like data sparsity and generalization in computational drug design.

7 Conclusion

DiffDock represents a significant advancement in molecular docking by leveraging diffusion generative modeling to predict ligand binding poses with high accuracy and robustness. Its ability to generate diverse and more realistic binding conformations, coupled with a practical confidence scoring mechanism, addresses several limitations of traditional and deep learning-based docking approaches.

Despite its strengths, the method's dependence on computational resources poses challenges for synthesizability, as demonstrated in our attempts to reproduce its results. Nevertheless, DiffDock's strong correlation between confidence scores and binding affinities underscores its utility in drug discovery workflows.

Looking ahead, integrating considerations for receptor flexibility, chemical feasibility of predicted ligands, and computational optimizations could further enhance its practicality and impact. Overall, DiffDock sets a strong foundation for future work in molecular docking, protein-protein interactions, and drug design, offering a promising direction for advancing computational methods in structural biology and pharmaceutical research.

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